

Linux & Git Cheat Sheet

Basic Linux commands • Basic Git commands • Git's three-tree architecture

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BASIC LINUX COMMANDS

Command	What it does	Sample usage
ls -la	List all files (incl. hidden), long format	ls -la ~/Downloads
cd ~/projects	Change into a directory	cd ~/projects/itec111
pwd	Print current working directory	pwd
mkdir -p a/b	Create nested directories	mkdir -p site/css/img
rm -rf build/	Delete a folder recursively	rm -rf node_modules/
cp -r src dst	Copy a folder and its contents	cp -r template new_site
mv old.txt new.txt	Rename or move a file	mv draft.md README.md
cat notes.txt	Print a file's contents	cat /etc/os-release
less log.txt	View a file page by page	less /var/log/syslog
touch file.txt	Create an empty file	touch index.html
find . -name '*.py'	Search for files by name	find . -name '*.py'
grep -i 'err' log	Search text (case-insensitive)	grep -rn 'TODO' src/
chmod +x run.sh	Make a script executable	chmod +x deploy.sh
chown user:grp f	Change file owner/group	sudo chown jay:jay app.py
man ls	Open manual page for a command	man grep
ps aux	List running processes	ps aux grep python
kill -9 <pid>	Force-stop a process	kill -9 4821
df -h	Show disk space (human-readable)	df -h /home
du -sh folder/	Show folder size	du -sh ~/Videos
tar -czvf a.tgz d/	Compress a folder to .tgz	tar -czvf backup.tgz site/

USEFUL PIPES, REDIRECTION & EXTRAS

Command	What it does	Sample usage
cmd1 cmd2	Pipe output of cmd1 into cmd2	ps aux grep node
ls > out.txt	Redirect output to a new file	ls -la > files.txt
echo hi >> log.txt	Append output to a file	date >> activity.log
cmd 2>&1	Redirect errors to standard output	build.sh > out.log 2>&1
sort file.txt	Sort lines alphabetically	sort names.txt
wc -l file.txt	Count lines in a file	wc -l students.csv
xargs rm	Build/run a command from input	find . -name '*.tmp' xargs rm
alias ll='ls -la'	Create a command shortcut	alias gs='git status'
export VAR=val	Set an environment variable	export PATH=\$PATH:~/bin
sudo apt update	Run a command as administrator	sudo apt install git

GIT'S THREE-TREE ARCHITECTURE

- **Working Directory** – the actual files on disk that you edit. Changes here are "unstaged" until Git is told about them.
- **Index / Staging Area** – a snapshot-in-progress. git add copies changes from the working directory into the index, preparing the next commit. It lets you commit only part of your changes.
- **HEAD** – a pointer to the last commit on the current branch; it represents the project's last saved snapshot. git commit takes what's in the index and records it permanently, moving HEAD forward.

edit (Working Dir) → git add → stage (Index) → git commit → save (HEAD)

git status compares all three trees at once; git diff compares Working Dir vs. Index; git diff --staged compares Index vs. HEAD.

git reset --soft moves HEAD only; --mixed moves HEAD + Index; --hard moves all three trees.

Sample: echo "hi" > a.txt && git add a.txt && git commit -m "add a.txt" moves a.txt through all three trees.

BASIC GIT COMMANDS

SETUP & SNAPSHOTS

Command	What it does	Sample usage
git init	Turn the current folder into a Git repo	git init my-site
git clone <url>	Copy a remote repository locally	git clone https://github.com/jay/site.git
git status	Show changed/staged/untracked files	git status
git add file.py	Stage one file for the next commit	git add app.py
git add -A	Stage all changes in the repo	git add -A
git commit -m 'msg'	Save staged changes as a snapshot	git commit -m 'Add navbar'
git commit -am 'msg'	Stage tracked files + commit in one step	git commit -am 'Fix typo'

HISTORY & INSPECTION

Command	What it does	Sample usage
git log --oneline	Compact commit history	git log --oneline -5
git diff	Unstaged changes vs. the index	git diff app.py
git show <commit>	Show details of one commit	git show a1b2c3d
git blame file.py	Show who last changed each line	git blame app.py

BRANCHING & MERGING

Command	What it does	Sample usage
git branch	List branches	git branch
git switch -c new	Create + switch to a new branch	git switch -c feature-login
git checkout main	Switch to an existing branch	git checkout main
git merge feature	Merge a branch into the current one	git merge feature-login
git rebase main	Replay commits on top of another branch	git rebase main

UNDOING & REMOTES

Command	What it does	Sample usage
git restore file.py	Discard unstaged changes in a file	git restore app.py
git reset --soft HEAD~1	Undo last commit, keep changes staged	git reset --soft HEAD~1
git revert <commit>	Create a new commit that undoes one	git revert a1b2c3d
git stash	Temporarily shelve uncommitted changes	git stash && git stash pop
git remote -v	List remote repositories	git remote -v
git fetch	Download remote changes, don't merge	git fetch origin
git pull	Fetch + merge into current branch	git pull origin main
git push -u origin main	Upload commits, set upstream tracking	git push -u origin main
git tag v1.0	Mark a commit with a version label	git tag v1.0
git config --global user.name 'x'	Set your committer name globally	git config --global user.name 'Jay'

TYPICAL DAILY WORKFLOW EXAMPLE

- git pull — get the latest changes from the team.
- Edit files in the Working Directory (e.g. edit index.html).
- git status — check what changed.
- git add . — stage the changes you want to save.
- git commit -m "message" — e.g. git commit -m "Update homepage layout".
- git push — upload your commits, e.g. git push origin main.

SAMPLE .GITIGNORE ENTRIES

Command	What it does	Sample usage
__pycache__/	Ignore Python cache folders	__pycache__/
node_modules/	Ignore installed Node packages	node_modules/
*.log	Ignore all log files	debug.log

REFERENCES

- Git. (n.d.). Git - Reset Demystified (the three trees). Git Documentation. <https://git-scm.com/book/en/v2/Git-Tools-Reset-Demystified>
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- GNU Project. (n.d.). Core Utilities Manual (ls, cp, mv, rm, find, du, df, etc). <https://www.gnu.org/software/coreutils/manual/coreutils.html>

"The author used Claude (Anthropic) to help organize and format this cheat sheet during assignment preparation. The author reviewed and edited the content as needed and takes full responsibility for the final content of the work."